From local quantum-field measurements to certified resources

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Abstract

Localized quantum probes connect quantum-field correlations with finite systems whose properties can be measured. This connection raises a foundational question: which assumptions allow a local measurement to support a claim of resource extraction? We critically review the inference from admissible field operations to probe states, resource criteria, and observable records. Three distinctions organize the discussion: causal admissibility versus apparatus realization, field correlations versus detector resources, and a model prediction versus an experimentally certified property. Detector instruments, algebraic system–probe schemes, and time-resolved event descriptions are compared by the objects they determine. Data processing constrains accessible correlations only after the relevant field algebras and complete interaction supports have been specified. Finite-dimensional counterexamples show why residual negativity need not certify mixed-state multipartite entanglement and why protection under a local channel need not identify a relativistic mechanism. Recent contextuality results illustrate the dependence of a resource claim on preparation and measurement scenarios. Composite observables and a finite-sample witness bound clarify separate requirements for dynamical, ultraviolet, calibration, and sampling control. Communication and metrology enter as conditional tests of utility. The resulting synthesis identifies what harvesting calculations establish and what additional evidence is required for a reliable resource claim.

Keywords: Quantum field theory, Local measurements, Relativistic causality, Entanglement harvesting, Quantum resources, Resource certification

1 Introduction

A measurement in quantum field theory must connect an extended physical interaction with a localized record. This requirement brings together relativistic causality, the dynamics of an apparatus, and the interpretation of a quantum state. Commuting observables in spacelike separated regions do not by themselves specify which measurements a laboratory can perform, how a record is produced, or which state update is warranted. These questions are central to both the foundations of quantum field theory and relativistic quantum information [1–3].

Resource-harvesting protocols add a further question. Initially uncorrelated probes can, in specified models, become entangled through local interactions with a correlated field [4, 5]. The field correlation, the reduced probe state, and the statistics of a probe measurement are nevertheless distinct objects. A positive expression computed from a truncated state establishes a different kind of claim from an entanglement witness measured with controlled uncertainty. Even an experimentally certified probe resource requires additional assumptions before its origin can be attributed to pre-existing field correlations.

This review asks: *which assumptions allow a local quantum-field measurement to support a claim of resource extraction?* We address this question by following the inference from a field state and admissible interaction to a probe state, a resource criterion, and a measurement record. Three distinctions organize the analysis: causal admissibility and apparatus realization; field correlations and detector resources; and a theoretical prediction and an experimentally certified property. Communication and parameter estimation enter at the end of this chain, as tasks whose performance requires its own specification.

The closest measurement-theory synthesis is the review by Papageorgiou and Fraser [3]. It organizes detector-based, algebraic, and histories-based proposals around Sorkin-type impossible measurements, and examines the roles of dynamics, local algebras, and state updates. We build on those distinctions. Our organizing question concerns the further inference to a resource claim once an operation or a record has been specified. In particular, agreement of outcome probabilities need not establish agreement of back-action; a region assigned to a detector channel must contain its complete interaction support; and a resource criterion must refer to the preparation and measurement scenario actually used. This emphasis complements the earlier review rather than replacing its account of the local-measurement problem.

Detector and atom–field reviews provide the dynamical background, while quantum-resource reviews supply the relevant state criteria and conversion rules [2, 6–8]. We use elementary instrument examples, data-processing inequalities, and finite-state counterexamples to expose assumptions needed when connecting these bodies of work. The inequalities are not an optimal harvesting construction, and the counterexamples are not a new resource theory. Their purpose is to identify which inferences are valid and which need an additional dynamical, operational, or statistical premise.

The scope is a critical narrative review of selected foundational and recent work through 4 November 2026. Selection follows relevance to localized field operations,

probe resources, and their certification, using topic and title searches, publication-metadata checks, and targeted reading of directly relevant sources. The review is not an exhaustive history of relativistic quantum information or a systematic-review protocol. Preprints are identified explicitly. Background-specific spectra, complete radial scattering calculations, and quantum-network engineering are outside the present scope. The supplementary material retains only derivations and numerical diagnostics that support the stated inference problem; it distinguishes exact identities, perturbative expressions, and numerical convergence checks.

Section 2 relates outcome probabilities, instruments, and causal realization. Section 3 identifies the field regions and channel assumptions entering accessibility bounds. Section 4 separates harvesting from signal-mediated generation. Section 5 examines scenario-dependent resource claims and mixed-state diagnostics. Section 6 treats the coupled observable, approximation control, and readout. Section 7 explains why certification and task performance remain distinct, and Sec. 8 states the resulting limits on interpretation.

2 Local operations, records, and causal realization

2.1 The objects entering a resource claim

We use $c = \hbar = k_B = 1$ and metric signature $(-,+,+,+)$. For a real scalar field in state ρ_ϕ, define

$$W(x,x') = \mathrm{Tr}[\rho_\phi \hat{\phi}(x)\hat{\phi}(x')]. \tag{1}$$

This two-point function describes correlations. For spacelike separated x, x', microcausality instead requires

$$[\hat{\phi}(x), \hat{\phi}(x')] = 0. \tag{2}$$

In a free scalar theory, the commutator is a state-independent c-number distribution related to the causal propagator. Nonzero spacelike correlations and vanishing causal response are consequently compatible. Whether a particular probe accesses those correlations depends on its interaction and preparation.

The relevant inference has the form

$$\begin{aligned} \text{field state} &\longrightarrow \text{localized interaction} \\ &\longrightarrow \text{record or probe state} \\ &\longrightarrow \text{resource criterion} \longrightarrow \text{task}. \end{aligned} \tag{3}$$

A local field algebra, a finite probe density matrix, and a classical record occupy different positions in this chain. A resource criterion additionally specifies a subsystem, a set of free states, and, where a monotone is claimed, an allowed operation class. The distinctions are needed even when two calculations use the same field correlator.

2.2 Probabilities, instruments, and local apparatus

Applying an ideal projection postulate to an arbitrary QFT observable can produce Sorkin-type impossible-measurement scenarios. The issue is which operations a

Table 1 Different claims made in a local-probe experiment. The rows concern different inference targets; no row follows solely from the preceding one. The table is an assessment scheme, not a new resource theory.

Claim	What must be specified
Field correlations	State and local algebra or regulated subsystem.
Accessible probe state	Initial probe resources, coupling, complete supports, and resulting channel.
Resource certification	Resource scenario, readout observables, calibration and statistical control.
Harvesting attribution	Initial product assumptions and exclusion or quantification of inter-probe causal influence.
Operational utility	Encoding or estimator, allowed access, cost budget and attainable performance.

spacetime-localized apparatus can implement, not simply whether two observables commute [3, 9, 10]. A POVM specifies outcome probabilities; an instrument additionally specifies conditional state changes [11, 12]. Two constructions can agree on a probability without inducing the same back-action.

A two-level Unruh–DeWitt (UDW) detector provides a tractable apparatus model. For a worldline $x(\tau)$,

$$H_I(\tau) = \lambda\chi(\tau)\mu(\tau)\hat{\phi}[x(\tau)]. \tag{4}$$

Here λ is the coupling constant, χ is a real switching function, and $\mu(\tau) = \sigma^+ e^{i\Omega\tau} + \sigma^- e^{-i\Omega\tau}$ is the monopole operator. The proper energy gap is $\Omega > 0$, with $\sigma^+ = |e\rangle\langle g|$ and $\sigma^- = (\sigma^+)^\dagger$ for excited and ground states $|e\rangle, |g\rangle$. Finite-size detectors replace the point evaluation by a spatial smearing. The preparation, gap, switching, profile, and readout determine the induced field measurement and the final probe state. Detector-based measurement theory makes that induced POVM and update explicit [13]. Fully relativistic localized-field probes reproduce leading detector predictions in specified regimes, supporting the approximation without identifying all probe models or perturbative orders [14–16].

The Fewster–Verch framework instead couples target and probe QFTs in a compact region and induces observables and instruments through a scattering morphism [17]. Local couplings preserve causal consistency in the system–probe setting [18]. Constructive results for linear scalar fields include asymptotic local-probe schemes and specified smeared-field measurements [19, 20]. Their assumptions do not amount to a classification of every instrument in interacting QFT.

The distinction between a causal map and a specified realization is now sharper. In a free Klein–Gordon theory, Mandrysch, Simmons, and Navascués construct causal Weyl channels that cannot be approximated arbitrarily well by the FV schemes considered, while proving realizability for a broad class of diagonal Weyl instruments [21]. This preprint separates two proposed classes of operations; it does not establish that every causal operation is physical, or that FV schemes exhaust all physical apparatus. Its heralded constructions must also be assessed together with the success probability

and the nonselective instrument. For harvesting, a vanishing signalling test is therefore one requirement: a claim about an attainable probe resource still needs an explicit coupling, apparatus state, and readout.

2.3 Equal probabilities can accompany different back-action

An induced instrument on the field and an output channel to retained probes are related through a specified system–apparatus realization. An abstract field instrument need not fix which apparatus degrees of freedom are retained. A probe resource must therefore be attached to an identified output subsystem and preparation, rather than to a field POVM or update alone.

An elementary finite-dimensional example shows why outcome agreement is insufficient. Let S be a qubit, $r \in \{0, 1\}$ a record, and define two instruments

$$\mathcal{I}_r(\rho) = \tfrac{1}{2}\rho, \qquad \mathcal{J}_r(\rho) = \tfrac{1}{2}\,\mathrm{Tr}(\rho)|0\rangle\langle 0|. \tag{5}$$

Both give $p_r = 1/2$ for every normalized input and have the same effects $E_r = I/2$. Their nonselective channels are respectively the identity and replacement by $|0\rangle$. If S initially shares a Bell state with an untouched reference, the first channel preserves that entanglement and the second removes it. Both operations are local on S. Thus a common outcome distribution does not determine a common resource-bearing output. Supplementary Sec. S1 supplies the Kraus representation and the explicit conditional states. This example concerns back-action on a pre-existing resource, rather than a field-harvesting protocol.

Selective state updating must also be distinguished from signalling. For a local instrument $\{\mathcal{I}_r\}$ on S, with a trace-preserving nonselective channel and an untouched system B,

$$\sum_r p_r \rho_{B|r} = \rho_B. \tag{6}$$

The conditional state $\rho_{B|r}$ is defined for $p_r > 0$. The identity follows by tracing over S and using trace preservation. A branch may differ from the initial marginal, but identifying the branch requires the record. In QFT one must establish the corresponding localization and causal-composition properties of the instrument; commutation of a pair of field observables is not a substitute for that construction.

2.4 Event probabilities and causal composition

Quantum temporal probabilities (QTP) retain detection time as an outcome. Their leading event density has the schematic form

$$P(x, r) = \int d^4y\, R^{ab}(y, r) G_{ab}(x - y/2, x + y/2), \tag{7}$$

where x is the coarse-grained event location, r labels the recorded outcome, y is the relative spacetime coordinate, and the component indices a, b are contracted. The field correlator is G_{ab}; R^{ab} describes apparatus response and coarse-graining. Joint-event probabilities involve selected unequal-time higher-point correlators. UDW

response and Glauber photodetection arise in limiting regimes, rather than defining interchangeable measurements [22–25]. Event-hierarchy proposals examine quantum features of multi-time probabilities; these indicators concern a specified hierarchy, not automatically the entanglement of a reduced detector state [26].

Other approaches emphasize the composition of localized operations. Positive-formalism constructions address positivity and composition for specified observable classes; a multi-observable expectation is not itself a joint outcome distribution [27, 28]. Local S-matrix constructions test causal factorization and stronger conditions on operations, including pointer back-action in a specified model [29]. These results complement explicit detector instruments. Their different output objects are summarized in Table 2.

Table 2 Operational outputs of complementary measurement constructions.

Construction	Direct output	Additional requirement for a resource claim
Specified detector model [13]	Probe probabilities, induced field update, final probe state	Controlled dynamics, readout model, and a witness on that state.
FV system–probe scheme [17]	Induced local observables and instruments	A specified probe preparation and implementation of the relevant readout.
QTP [24]	Time-resolved and joint event statistics	An explicit link between those statistics and the proposed resource criterion.
Positive formalism / local S matrix [28, 29]	Probabilistic composition / admissible local operations	A matching apparatus and outcome functional; causal admissibility alone does not specify detector negativity.

A useful benchmark fixes one field state, compact interaction supports, detector preparation, smearing, switching, and pointer readout. It first compares integrated single-probe probabilities, then joint event statistics and selective updates. The final detector state requires a specified probe output channel or a joint system–apparatus realization, including the retained probe subsystem. Supplementary Sec. S1 gives this benchmark, including the distinction between coincidence covariance and negativity. It is a proposed comparison protocol, not an already completed equivalence proof between the frameworks.

3 Field regions and accessible probe resources

3.1 A data-processing bound on detector-accessible entanglement

The field-to-detector distinction can be quantified without assuming a spatial tensor-product decomposition of the field. Let $\mathcal{A}_A$ and $\mathcal{A}_B$ be commuting local algebras for spacelike separated regions, and assume that a normal product reference state exists (for example, under a suitable split or finite-resolution prescription). Define

$$I_{\mathrm{F}}(A:B) = S(\omega|_{\mathcal{A}_A\vee\mathcal{A}_B} \,\|\, \omega|_{\mathcal{A}_A} \otimes \omega|_{\mathcal{A}_B}) \,. \tag{8}$$

Here ω denotes the field state and $\mathcal{A}_A \vee \mathcal{A}_B$ the algebra generated by the two local algebras; S is Araki relative entropy. For finite-dimensional density matrices, $S(\rho\|\sigma) = \mathrm{Tr}[\rho(\ln\rho - \ln\sigma)]$ when $\mathrm{supp}\,\rho \subseteq \mathrm{supp}\,\sigma$, and is infinite otherwise [30]. In gauge theories, the local algebras, shared centre, and superselection sectors must be specified first. The Araki relative entropy may still be infinite; modular p-nuclearity for some $0 < p < 1$ is a sufficient finiteness condition in the class studied by Panebianco and Wegener [31]. Thus the following bound is quantitatively informative only when $I_{\mathrm{F}} < \infty$, and regulated results must state their regulator and limiting procedure.

Suppose D_A and D_B interact locally with $\mathcal{A}_A$ and $\mathcal{A}_B$, respectively. Initially the probes are uncorrelated with the field and with one another, and share no nonlocal ancilla or direct interaction. For the final probe state, $I(D_A:D_B) = S(\rho_{AB}\|\rho_A \otimes \rho_B)$, where ρ_A, ρ_B are its marginals. The induced local channels and monotonicity of relative entropy give

$$I(D_A:D_B) \le I_{\mathrm{F}}(A:B). \tag{9}$$

For the detector state, define the relative entropy of entanglement

$$E_R(\rho_{AB}) = \inf_{\sigma\in\mathrm{SEP}} S(\rho_{AB}\|\sigma). \tag{10}$$

Here SEP is the set of separable probe states. Since $\rho_A \otimes \rho_B$ is separable, $E_R(\rho_{AB}) \le I(D_A:D_B)$. Let d_A, d_B be the probe Hilbert-space dimensions, T_B partial transposition on B in a fixed basis, and $\|\cdot\|_1$ the trace norm. Quantum Pinsker's inequality (with natural logarithms) and $\|X^{T_B}\|_1 \le d_B\|X\|_1$ then imply

$$\mathcal{N}(\rho_{AB}) \le d_B\sqrt{\frac{E_R(\rho_{AB})}{2}}, \tag{11}$$

where $\mathcal{N} = (\|\rho_{AB}^{T_B}\|_1 - 1)/2$. All entropies in these bounds use natural logarithms; base-two entropies require the corresponding factor of $\ln 2$. Combining the bounds yields

$$\mathcal{N}(\rho_{AB}) \le \min\left\{\frac{\min(d_A,d_B)-1}{2}, d_B\sqrt{\frac{I_{\mathrm{F}}(A:B)}{2}}\right\}, \tag{12}$$

or, for two qubits,

$$\mathcal{N}(\rho_{AB}) \leq \min\left\{\frac{1}{2}, \sqrt{2I_{\mathrm{F}}(A:B)}\right\}. \tag{13}$$

The inequality is rigorous but generally not tight; it follows from data processing and finite-dimensional norm bounds, not from an optimal harvesting construction.

To obtain the intermediate estimate, choose a separable σ approaching the infimum in E_R. Pinsker's inequality gives $\|\rho - \sigma\|_1 \leq \sqrt{2S(\rho\|\sigma)}$. Since $\|\sigma^{T_B}\|_1 = 1$ and partial transposition obeys $\|X^{T_B}\|_1 \leq d_B\|X\|_1$, the triangle inequality yields $\mathcal{N}(\rho) \leq (d_B/2)\|\rho-\sigma\|_1$. Taking the infimum gives Eq. (11). The dimension-dependent cap in Eq. (12) is the maximum possible negativity for the detector dimensions.

No positive lower bound on detector entanglement follows from I_{F} alone: a probe may miss the correlated modes, have zero coupling, or induce an entanglement-breaking channel. In a specified perturbative Unruh–DeWitt model, the condition $|\mathcal{M}| - \sqrt{P_A P_B} > 0$ is detector-specific, not a field-only lower bound. Local noise can only reduce the unconditioned detector correlations; a postselected branch must be weighted by its success probability when assessing average yield.

Three levels must be distinguished. Field mutual information $I_{\mathrm{F}}(A : B)$ measures total correlations between local field algebras, including classical and quantum parts. Detector mutual information $I(D_A : D_B)$ measures total correlations in the final probe state and depends on the chosen preparation and local channels. Detector entanglement, quantified here by E_R or negativity, is more restrictive and is not the entanglement of the full field regions. The data-processing and separable-state bounds relate these quantities but do not make them equal: $I_{\mathrm{F}} > 0$ does not guarantee nonzero detector negativity.

A complementary Gaussian-mode result bounds the entanglement extractable by a fixed number of localized bosonic probes using the most mixed normal modes of the reduced covariance matrix [32]. It concerns mode selection in a pure Gaussian state, not algebraic mutual information or a general detector-channel bound.

3.2 Channel dependence and finite-resource bounds

Data processing is universal, whereas a strict contraction factor is channel dependent. For local probe channels $\mathcal{E}_A, \mathcal{E}_B$, a bound of the form

$$I(D_A : D_B) \leq \eta_A \eta_B I_{\mathrm{F}}(A : B), \qquad 0 \leq \eta_A, \eta_B \leq 1, \tag{14}$$

is useful only if the coefficients are established for the same input class and regulator as the field mutual information. Complete contraction and quantum Doeblin methods give sufficient conditions in specified noisy-channel settings [33, 34]. They do not assign a detector-independent extraction efficiency to a relativistic field.

The ideal Jaynes–Cummings channel illustrates the obstruction: rare, high-energy Fock inputs can preserve an encoded bit almost perfectly even under a fixed mean-energy budget. An average-energy constraint need not imply a strictly smaller worst-case mutual-information contraction, a mechanism related to classical average-cost counterexamples [35]. A hard cutoff, explicitly calibrated noise, or a state-dependent one-sided data-processing estimate can be more informative. Supplementary Sec. S2

gives the detailed argument; the finite-mode transfer example in Supplementary Sec. S2 illustrates the difference between field mutual information, detector mutual information, and detector negativity [36]. These examples explain the limits of an upper bound; they are not continuum harvesting protocols.

3.3 The pointlike limit of strictly spacelike detectors

A compact-support calculation for two identical static UDW detectors in the four-dimensional massless scalar vacuum has a strictly positive leading-order entangling gap in the pointlike limit when the switching duration and detector separation are fixed. Let $2T$ be the switching duration, L the centre separation, and σ the spatial support radius. The complete interaction supports obey $L - 2\sigma > 2T$ throughout the scan. Define $P_\sigma = \lambda^2 p_\sigma$ and $\mathcal{M}_\sigma = \lambda^2 m_\sigma$ at leading order. At $T = 1$, $\Omega T = 9$, and $L/T = 2.033333\ldots$, the pointlike response coefficients are $p_0 = 2.13325790 \times 10^{-6}$ and $|m_0| = 3.57653182 \times 10^{-6}$, giving the second-order result $\lim_{\sigma\to 0} \mathcal{N}_\sigma^{(2)}/\lambda^2 = 1.44327392 \times 10^{-6}$. Thus the positive leading-order entangling gap is not caused by a finite spatial detector radius. This conclusion concerns the detector state for the specified field, switching, and weak-coupling regime; it does not determine the mutual information of shrinking equal-time field regions. The full compact profiles, small-size expansion, quadrature checks, and figure are retained in Supplementary Sec. S3.

3.4 Which field region controls the data-processing bound?

The pointlike detector limit raises a geometric issue in applying the field-to-detector data-processing bound. The detector interaction is not supported on an equal-time ball of radius σ. It is supported on the spacetime world tube

$$\mathcal{W}_A = \{(t, \mathbf{x}) : |t| < T,\ |\mathbf{x} - \mathbf{x}_A| < \sigma\}, \tag{15}$$

and similarly for detector B. A centred causal double cone containing this nominal open world tube is the domain of dependence of the equal-time ball of radius

$$R = T + \sigma. \tag{16}$$

Indeed, the double cone $D(B_R)$ is characterized by $|t| + |\mathbf{x} - \mathbf{x}_A| < R$, which contains every point of $\mathcal{W}_A$ when $R \geq T + \sigma$. For a compact interaction support inside this nominal open tube, R suffices. To include the closed support of the profiles used in the benchmark, take $R_\epsilon = T + \sigma + \epsilon$, with $0 < \epsilon < L/2 - T - \sigma$. We use $R = T + \sigma$ as the limiting enclosing radius. The strict-spacelike condition $L - 2\sigma > 2T$ is exactly $L > 2R$. Thus the two enclosing double cones remain spacelike separated throughout the pointlike limit, with $R \to T$ rather than $R \to 0$.

The local detector channels are therefore localized in the algebras of enclosing double cones, with the small enlargement when needed, rather than those of equal-time balls of radius σ. Under the split/finite-resolution assumptions used above, the appropriate data-processing statement is

$$I(D_A : D_B) \leq I_{\mathrm{F}}(D(B_{R_\epsilon})_A : D(B_{R_\epsilon})_B)\,. \tag{17}$$

By the time-slice property, the algebra of a causal double cone can be represented on its equal-time Cauchy ball. This permits the use of spatial-ball mutual-information methods, with radius R_ϵ, rather than σ.

This distinction matters because the known large-separation expansion for two equal-time balls in the four-dimensional massless scalar vacuum has the leading scaling

$$I_{\mathrm{F}}(B_R^A : B_R^B) \sim C_{\mathrm{sc}} \left(\frac{R}{L}\right)^4, \qquad R/L \ll 1, \tag{18}$$

where C_{sc} depends on the field theory and the region convention [37, 38]. This is the small-region expansion of the field mutual information; it is not the scaling of the detector negativity. For a fixed admissible $\epsilon > 0$, substituting the enclosing radius gives

$$I_{\mathrm{F}} \sim C_{\mathrm{sc}} \left(\frac{T+\sigma+\epsilon}{L}\right)^4 \longrightarrow C_{\mathrm{sc}} \left(\frac{T+\epsilon}{L}\right)^4 \quad (\sigma \to 0), \tag{19}$$

provided the large-separation regime applies. The field mutual-information budget therefore does not vanish merely because the detector's spatial radius tends to zero while its interaction duration remains fixed.

For the numerical harvesting point used above, $L/T = 2.033333\ldots$ and the limiting enclosing radius is $R/T = 1$ as $\sigma, \epsilon \to 0$. The equal-sphere cross ratio is $z = 4R^2/L^2 \simeq 0.9675$, so the large-separation expansion in Eq. (18) is not quantitatively controlled there. It would be incorrect to insert that asymptotic formula as a numerical value for the bound. A direct Gaussian-field or regulated algebraic calculation of the mutual information of the two nearly touching double cones is required.

The geometric conclusion does not require a numerical mutual-information ansatz: assigning a fixed-duration interaction to shrinking equal-time balls changes the subsystem being bounded. A quantitative field-derived negativity bound still needs a controlled value or rigorous upper bound for the mutual information of regions containing the full interaction supports.

4 Entanglement harvesting and causal attribution

4.1 The detector-state criterion

Harvesting couples initially uncorrelated probes locally to a field and tests their final reduced state. Vacuum-extraction proposals and detector calculations established that pre-existing field correlations can be accessed without signal exchange between the complete interaction regions [4, 5, 39]. For initially ground-state two-level probes with vanishing one-point function of the coupled field observable, let P_A, P_B denote the second-order excitation probabilities and $\mathcal{M}$ the ground–double-excitation coherence. The relevant second-order partial-transpose block has diagonal entries P_A, P_B and off-diagonal entries $\mathcal{M}, \mathcal{M}^*$. Its potentially negative eigenvalue is

$$\lambda_-^{(2)} = \frac{P_A + P_B - \sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2}}{2}. \tag{20}$$

With $\mathcal{N} = (\|\rho^{T_B}\|_1 - 1)/2$, the leading negativity is therefore

$$\mathcal{N}^{(2)} = \max\left\{0, \frac{\sqrt{(P_A - P_B)^2 + 4|\mathcal{M}|^2} - (P_A + P_B)}{2}\right\}. \tag{21}$$

The condition $|\mathcal{M}| > \sqrt{P_A P_B}$ is a threshold, not the general value of negativity. For identical probes it reduces to $\max(0, |\mathcal{M}| - P)$. The local probabilities and nonlocal coherence must be evaluated at the same order; omitted terms can decide the sign near threshold.

The quantities are interaction dependent. Switching, smearing, derivative coupling, detector dimension, and initial preparation change the sampled kernels. Antiparallel and parallel acceleration can consequently produce different trends in specified setups [40]. Localized relativistic QFT probes reproduce leading harvesting predictions under stated assumptions [16]; Gaussian-mode extraction bounds instead concern mode selection in a specified covariance-matrix setting [32]. Neither establishes a universal detector-independent harvesting yield.

4.2 Full support and the origin of the nonlocal amplitude

For strictly spacelike interaction supports, microcausality excludes inter-probe signalling. If the supports are causally connected, the final entanglement can contain a signal-mediated contribution. For free scalar models at leading order, a commutator–anticommutator decomposition separates state-independent causal response from state-dependent correlations [41]. The amplitudes can interfere:

$$|\mathcal{M}|^2 = |\mathcal{M}_{\rm corr}|^2 + |\mathcal{M}_{\rm comm}|^2 + 2\,\mathrm{Re}(\mathcal{M}_{\rm corr}\mathcal{M}^*_{\rm comm}). \tag{22}$$

Negativity is nonlinear in the total amplitude and local noise. It cannot in general be partitioned into two positive independent yields. Constructive and destructive interference are model-dependent possibilities [42, 43].

Gaussian profiles have noncompact tails. Centre separation or a small communication-mediated-entanglement estimator does not prove exact spacelike separation of the interactions. Compact support establishes the strict condition; a tail approximation instead requires an explicit error or signalling budget. The compact benchmark in Sec. 3.3 supplies an example of the former. Covariance of a specified perturbative calculation and absence of signal exchange are separate checks [44].

4.3 Optimization and finite-resource interpretation

Optimizing temporal profiles can improve the leading-order harvesting gap, but the optimized profile must remain inside the approximation regime. The published Hermite-basis optimization fixes the L^2 norm of the switching relative to a Gaussian reference and distinguishes different signalling constraints [45]. This normalization removes an arbitrary overall rescaling of the interaction; it does not fix peak drive, control bandwidth, or laboratory work. Approximately compact profiles also retain tails. Large gains in a second-order coefficient therefore require separate checks of causal

Table 3 Causal classifications and the evidence they require.

Regime	Required check	Interpretation
Strict spacelike extraction	Every pair of points in the two compact interaction supports is spacelike	Inter-probe signal exchange is excluded; the probe-state witness still needs approximation control.
Effective separation with tails	Tail contribution or signalling error is bounded for the stated profiles	Extraction is assessed within that error budget, without asserting exact spacelike support.
Causal contact	Full local and nonlocal amplitudes, including interference, are retained	A positive final witness alone does not distinguish harvesting from signal-mediated generation.
Postselected extraction	Conditional state and success probability are both specified	Conditional improvement must be distinguished from average resource yield.

leakage, control costs, and finite-coupling corrections. The authors explicitly identify the breakdown of second-order predictions when extrapolated to proposed implementations. Circuit-derived variable-gap detectors and Bose-polaron proposals introduce different microscopic constraints [46, 47]. Their relevance is the explicit map from controls to an effective interaction, not a common temperature or numerical witness value.

With detector ensembles, bipartite entanglement between two groups is not automatically genuine multipartite entanglement among all probes. Salomaa, Keski-Vakkuri, and Nadal-Gisbert obtain the leading bipartite negativity from a block whose dimension grows linearly with detector number, and find approximately linear growth in an alternating chain at fixed per-detector parameters [48]. This is a useful computational reduction and a specified perturbative scaling, rather than an unlimited gain at fixed total resources. Increasing detector number also changes the total interaction budget and may spoil uniform perturbative control. At minimum, the summed leading excitation probability $\sum_j P_j^{(2)}$ must remain small for a state expansion about the all-ground input; that diagnostic alone does not bound all higher-order terms. Access to additional field regions and classical records can also change the extractable resource [49]. Such protocols require those operations to be included in the resource budget.

Algebraic embezzlement addresses another limit: suitable local operations can borrow entanglement from a reference state with small state disturbance in the operator-algebraic setting [50, 51]. These results do not specify a finite-energy detector protocol with arbitrary yield. Here the operational quantity remains the resource produced by a stated probe interaction with controlled physical costs.

5 What a resource criterion certifies

5.1 A witness requires a measurement scenario

Entanglement, steering, Bell nonlocality, coherence, magic, and contextuality are defined using different operational structures. In a fixed bipartite scenario,

$$\text{Bell violation} \Longrightarrow \text{steerability} \Longrightarrow \text{entanglement}, \tag{23}$$

while the converses fail in general [52–55]. Coherence depends on a reference basis and free operations; its conversion to entanglement is a theorem within specified resource theories [56–59]. It does not identify detector coherence with negativity.

Magic–contextuality relations likewise depend on dimension, representation, and the allowed subtheory. Contextuality is necessary for computational advantage in the relevant odd-prime stabilizer setting, but arbitrary contextuality is not a sufficient distillation guarantee [60–62]. A contextual fraction refers to a specified empirical measurement model, including its compatible measurement contexts [63, 64]. Generalized contextuality instead tests ontological representations against operational equivalences among specified preparations, measurements, or transformations [65]. A conclusion about one such scenario is not a state-independent conclusion about all meanings of contextuality.

Table 4 Reference structures needed to interpret detector-resource studies.

Resource	Typical diagnostic	Information needed beyond its numerical value
Entanglement	Negativity, concurrence, entanglement witness	Bipartition, normalization, state reconstruction, and mixed-state criterion.
Steering / Bell nonlocality	Assemblage tests / Bell inequalities	Trusted parties, direction, measurement settings, and statistical confidence.
Coherence	Basis-dependent coherence or robustness	Reference basis, free operations, and conversion or discrimination task.
Magic / contextuality	Mana, stabilizer indicators / contextual fraction	Dimension, stabilizer subtheory, and the full measurement scenario.

Detector models make these requirements concrete. Nonidentical gaps can produce directional steering ranges [66]. Magic-harvesting studies use detector dimensions and stabilizer diagnostics that need not agree with entanglement ordering [67, 68]. LeMaitre uses a modified five-measurement pentagram scenario for a qutrit, whereas Lima *et al.* study compatible Heisenberg–Weyl measurements of two qutrits [69, 70]. Their contextual fractions refer to different empirical models and are not directly

interchangeable. Noncompact profiles also require causal-tail qualifications. The relevant comparison is the certified property under the actual measurement model, not a ranking across unrelated resource scales.

A preprint submitted on 30 October 2026, *No generalised contextuality from the vacuum*, makes this qualification consequential [71]. Its channel argument concerns a noncontextual *family of preparations* tested against all POVMs: an invertible CPTP map preserves the relevant operational equivalences, allowing an initial noncontextual model to be transported to the output family. Invertibility here is linear invertibility; the inverse need not be positive or physically implementable. The paper gives an exact gapless, Gaussian-field construction when the smeared variances are finite, and a weak-coupling argument near the identity channel. These statements do not refute contextual-fraction results in their specified qutrit measurement scenarios. They also do not establish a universal no-go theorem for arbitrary finite coupling, preparation families, singular smearings, or conditional postselected maps.

Supplementary Sec. S4 states the channel implication and its proof explicitly. In finite dimension, an exact channel continuous at zero coupling is invertible in a sufficiently small neighbourhood of the identity. By contrast, invertibility of each finite perturbative truncation does not prove invertibility of an arbitrary finite-coupling limit. A comparison must report the preparation family, measurement contexts or operational equivalences, allowed transformations, and any selection rule before calling an observed change "contextuality harvesting."

5.2 Residual negativity does not certify mixed-state GME

A mixed state is biseparable if it is a convex mixture of states separable across possibly different bipartitions. Genuine multipartite entanglement (GME) excludes every such decomposition. Residual combinations of bipartite negativities are not automatically faithful GME indicators on mixed states [72–74].

An explicit diagnostic is

$$\rho_{\mathrm{b}} = \frac{1}{3} \sum_{XY|ZW \in \{AB|CD, AC|BD, AD|BC\}} \Phi^{+}_{XY} \otimes \Phi^{+}_{ZW}, \qquad \Phi^{+} = |\Phi^{+}\rangle\langle\Phi^{+}|. \tag{24}$$

Here $|\Phi^{+}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Each component is separable across a $2|2$ cut, so the mixture is biseparable. Nevertheless, with the normalized convention $N = \|\rho^{T_A}\|_1 - 1 = 2\mathcal{N}$, all four $1|3$ negativities equal one and all six pair negativities vanish. Therefore the raw residuals

$$\pi_A = N^2_{A|BCD} - \sum_{X=B,C,D} N^2_{AX} \tag{25}$$

and the corresponding π_B, π_C, π_D all equal one. Their arithmetic mean $\pi_4 = (\pi_A + \pi_B + \pi_C + \pi_D)/4$ also equals one. A maximal value of this residual does not certify four-party GME. Supplementary Sec. S5 gives the three-party counterpart and the relation to the detector studies; the accompanying script directly diagonalizes the required partial transposes. We use "residual-negativity indicator" unless a mixed-state GME witness or an appropriate extension is separately established.

5.3 Structural invariance and perturbative revival are different claims

For

$$|\mathrm{CL}_4\rangle = \tfrac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle), \tag{26}$$

the reduced ABC state already has $N_{A|BC} = N_{B|AC} = 1$. Any local CPTP channel on D leaves that reduced state unchanged. Negativity monotonicity under tracing out D, together with the qubit upper cap, then implies $N_{A|BCD} = N_{B|ACD} = 1$ after the channel. Thus the corresponding freezing in the detector calculation [75] has an exact state-and-channel explanation, independent of a specifically Unruh-induced mechanism.

A different issue arises in finite-order revival plots. For a detector with proper acceleration $a > 0$ and gap Ω, set $q = e^{-2\pi\Omega/a}$ and let ν be the dimensionless effective coupling. In the first-order W and Dicke models considered in Supplementary Sec. S5, the normalization corrections are

$$\delta_W = \frac{\nu^2(1+3q)}{4(1-q)}, \qquad \delta_D = \frac{\nu^2(1+q)}{2(1-q)}. \tag{27}$$

The reconstructed turning points at the tested couplings occur with corrections of order unity [76, 77]. This demonstrates nonuniformity of those truncations as $q \to 1$, not absence of revival in the exact dynamics. A small normalization correction is a useful necessary diagnostic here, not a sufficient bound on all higher-order errors. Source-dependent reconstructions and independently testable state identities are distinguished in the supplement and verification record.

6 The coupled observable and the calibrated record

6.1 A common response template is not a universal spectrum

Expanding the interaction in Eq. (4) to second order, with $\hat{\phi}$ replaced by a Hermitian observable $\mathcal{O}$ and an initially ground-state probe uncorrelated with the field, gives the leading response

$$P_\nu^{(2)} = \lambda_\nu^2 \int d\tau\, d\tau'\, \chi_\nu(\tau)\chi_\nu(\tau') e^{-i\Omega_\nu(\tau-\tau')} W_{\mathcal{O}}(x_\nu(\tau), x_\nu(\tau')), \tag{28}$$

where $W_{\mathcal{O}}(x,x') = \langle \mathcal{O}(x)\mathcal{O}(x')\rangle$. Extended probes add spatial integrals. For two probes the same model must supply both local responses and the cross-correlator. Matching gaps or KMS temperatures does not match these kernels [78, 79].

Matching a stationary detailed-balance temperature does not determine the response normalization or the two-position cross-correlator. A finite-time interaction may also fail to reach the stationary regime. Background comparisons therefore require the coupled observable, state, trajectory, and complete switching to be fixed, in addition to a nominal temperature. Scalar mass and curvature coupling, tensor orientation, and composite-operator renormalization affect different parts of that specification [6, 80].

Table 5 Representative probe couplings and the controls needed for comparison.

Field	Coupled observable	Required controls
Real scalar	ϕ or a proper-time derivative	Mass, curvature coupling, state, infrared prescription, switching, and spatial profile.
Dirac	Even composite $:\bar{\psi}\psi:$ for the bosonic probe considered here	Spin connection, parallel transport, composite-operator prescription, and field state.
Maxwell	Electric dipole $d^i E_i$ or field strength	Gauge-invariant tensor kernel, tetrads, polarization, orientation, atomic form factors, and boundary conditions.
Composite scalar	$:\phi^2:$ or a specified renormalized energy density	One-point subtraction, connected correlations, distributional time ordering, spacetime dimension, and the full spatial and temporal profiles.

For a bosonic probe of a Dirac field, the even scalar density $:\bar{\psi}\psi:$ avoids a parity-changing linear coupling; its two-point function is a four-field correlator and needs its own prescription [81, 82]. An electromagnetic dipole couples through $d^i E_i$, where d^i and E_i are dipole and electric-field components in the chosen local frame. The tensor kernel, orientation, and atomic form factors then determine the sampled correlations [83]. Neither a field label nor a shared temperature fixes a harvesting yield.

For gauge theories, a gauge-invariant probe response and a regional entanglement assignment are separate constructions. The latter needs a local algebra and a treatment of centres, Gauss constraints, or boundary degrees of freedom [84, 85]. This is another reason to specify the subsystem before comparing a field-region quantity with a finite probe witness.

6.2 Composite observables: coherent means and ultraviolet control

The condition $\langle\phi\rangle = 0$ does not imply $\langle\mathcal{O}\rangle = 0$ for a quadratic observable. For example, a thermal state can have nonzero energy density after Minkowski-vacuum subtraction. Its mean acts as a local coherent drive and generates first-order probe coherences, so the unmodified ground-state block leading to Eq. (21) cannot simply be reused. For a prescribed mean, write $\mathcal{O} = \overline{\mathcal{O}}\, I + \delta\mathcal{O}$, where $\overline{\mathcal{O}}(x) = \langle\mathcal{O}(x)\rangle$. The appropriate leading fluctuation kernel is

$$W^{\mathrm{c}}_{\mathcal{O}}(x,x') = W_{\mathcal{O}}(x,x') - \overline{\mathcal{O}}(x)\overline{\mathcal{O}}(x'). \tag{29}$$

The mean-induced local evolution must be included consistently, or removed in a local interaction picture, before using a connected-kernel harvesting formula. This operation does not remove the physical energy cost of the local drive.

Renormalizing the observable's one-point function is also different from establishing that its time-ordered two-point function defines a finite detector calculation. Persistent divergences for quadratic couplings are known [86]. A recent preprint analyzes energy-density coupling and identifies switching-dependent coincident-time terms; for its Gaussian spatial profiles, smearing suffices in $1+1$ and $2+1$ dimensions but need not suffice in $3+1$ dimensions [87]. Smooth compact switchings with disjoint time supports remove the coincidence terms in the reductions studied there. This is a sufficient condition within the stated model, not a criterion for all composite observables or all detector geometries. In particular, the separated-detector cross-kernel must retain the spatial separation; a coincident-position pointlike reduction cannot stand in for it. Supplementary Sec. S6 gives an explicit separated-kernel check.

Ultraviolet finiteness and causal attribution are separate requirements. Temporally disjoint interactions may still be timelike related. Gaussian spatial profiles have overlapping tails, even if the probe centres are far apart. Conversely, a smooth kernel on two strictly spacelike compact supports has no light-cone coincidence singularity on their cross product. A reliable comparison must therefore specify the renormalization prescription, distributional pullback, full supports, and limiting procedure, rather than treating "smeared" or "normal ordered" as a complete regularity argument.

6.3 Certifying a measured probe state

A witness computed from an ideal reduced state certifies that model. Experimental certification also requires readout calibration and statistical uncertainty. If $p(r|\rho) = \mathrm{Tr}(E_r\rho)$, the map to observed counts includes detector preparation, coupling, switching, readout noise, and postselection. The sign convention matters. For a Hermitian entanglement witness W_{ent} satisfying $\mathrm{Tr}(W_{\mathrm{ent}}\sigma) \geq 0$ for all separable σ, entanglement requires a negative expectation with an *upper* confidence bound below zero. A resource monotone additionally obeys non-increase under a specified free-operation class; being a useful numerical indicator is not sufficient.

For example, write $W_{\mathrm{ent}} = w_0 I + \sum_{k=1}^{K} w_k O_k$, where the measured outcome of O_k lies in $[a_k, b_k]$. With n_k independent repetitions per fixed setting and total failure probability $\delta \in (0,1)$, Hoeffding's inequality and a union bound give the sufficient certificate [88]

$$\widehat{w} + \Delta_{\mathrm{stat}} + \Delta_{\mathrm{cal}} < 0, \qquad \Delta_{\mathrm{stat}} = \sum_{k=1}^{K} |w_k|(b_k - a_k)\sqrt{\frac{\ln(2K/\delta)}{2n_k}}. \tag{30}$$

Here $\widehat{w}$ is the measured witness estimate and Δ_{cal} is a separately justified bound on the systematic change in its expectation. The sampling model assumes a common prepared state and fixed measurement settings; drift, adaptive selection, or calibration fitted from the same counts requires an appropriate joint analysis. Supplementary Sec. S7 proves the bound and gives a Pauli-readout example. Unbounded optical quadratures require a justified tail or moment bound; Eq. (30) cannot be applied to them merely by clipping the data. Increasing the number of shots reduces Δ_{stat} but does not remove calibration bias or validate an approximate dynamical model.

Locality and detector reliability impose independent restrictions. Bounds relating sensitivity to vacuum dark counts depend on the apparatus and class of excitations considered [89]; they do not replace a resource-specific error analysis. Abrupt switching and finite smearing change ultraviolet behavior and the balance of local noise against nonlocal coherence [5, 90–92]. A prediction close to the resource threshold is especially vulnerable to omitted terms. Complete causal-support tests, numerical convergence, and control of omitted perturbative orders are therefore reported separately below.

6.4 Prediction, certification, and attribution require different evidence

A microscopic model is needed to predict a detector state, but a calibrated witness measured on the actual output can certify entanglement without assuming that a particular perturbative state is exact. Conversely, confidence in sampled counts does not control an omitted term in a theoretical expansion. These two uses of a witness should be kept separate.

For a theoretical estimate ρ_{mod}, suppose an independently justified trace-norm bound is available,

$$\|\rho - \rho_{\mathrm{mod}}\|_1 \leq \epsilon_{\mathrm{dyn}}. \tag{31}$$

For any bounded Hermitian witness W_{ent}, Hölder's inequality gives

$$|\operatorname{Tr}[W_{\mathrm{ent}}(\rho - \rho_{\mathrm{mod}})]| \leq \|W_{\mathrm{ent}}\|_\infty \epsilon_{\mathrm{dyn}}. \tag{32}$$

Here $\|\cdot\|_\infty$ is the operator norm. A predicted negative expectation survives the model uncertainty if its magnitude exceeds this bound. The usefulness of Eq. (32) depends on establishing ϵ_{dyn}; a small coupling or a positive numerical gap alone does not provide it. By contrast, Eq. (30) concerns sampling and measurement calibration for the actually prepared state.

Attributing an observed probe resource to the field is a further inference. Initial probe entanglement, a shared ancillary resource, a direct interaction, or causal leakage can change its origin without changing the fact that the measured state is entangled. A harvesting claim therefore combines state certification with the preparation and locality premises of the protocol. The same distinction explains why a conditional state change, which depends on a record, need not be a superluminal physical influence.

6.5 What the measured observable establishes

Electro-optic sampling (EOS) connects optical probes to terahertz field correlations. A detector-state entanglement witness has been proposed within this platform [93]. The vacuum/source-correlation experiment reported by Herter *et al.* resolves contributions associated with vacuum fluctuations and source radiation [94]. The measured correlations are an important field-sampling result; they do not by themselves certify entanglement of the reduced probe state or a Bell violation. The next inference requires the observables and calibrations of an appropriate probe-state witness.

Microscopically motivated detector proposals supply complementary evidence. Superconducting variable-gap models relate circuit controls to a derivative-coupled

detector interaction [46]. The Bose-polaron proposal maps trapped impurities to probes of an effective phonon field and estimates achievable witness strength and measurement demands [47]. These are implementation proposals. An analogue field and a relativistic vacuum are different physical inputs, even when part of the detector dynamics is reproduced.

The Bose-polaron example makes the certification bottleneck concrete. For one proposed operating point, the authors estimate local excitation probabilities of order 10^{-3} and a negativity of approximately 2×10^{-4}. Population measurements determine the local noise, while additional controlled rotations are required to infer the nonlocal coherence. Their analysis estimates readout infidelities below 10^{-5} and more than 10^5 experimental realizations to resolve the effect [47]. These are model-based feasibility estimates, not measured entanglement or a universal shot count. The signal is a difference between larger response terms, so both calibration and statistical precision matter. The phononic approximation, finite field temperature, control errors, and profile tails need their own error budgets; a small readout bias does not bound them. Equation (30) illustrates one conservative certification rule for bounded measurements, rather than reproducing the proposal's precision estimate.

Table 6 Evidence and the remaining step toward harvesting certification.

Platform / result	Evidence supplied	Additional requirement
EOS vacuum/source sampling [94]	Time-resolved optical field-correlation measurements	Probe-state entanglement witness with calibrated readout and uncertainty.
EOS harvesting design [93]	A platform-specific proposed witness	Experimental implementation and an entanglement-witness expectation whose upper confidence bound is below zero.
Variable-gap superconducting probes [46]	Circuit-derived interaction model and theoretical dynamics	Device calibration, controlled approximations, and a causal-support or signalling-error budget.
Bose-polaron probes [47]	Effective-field mapping and feasibility estimates	Validity of the mapping, loss and sampling analysis, and measured nonlocal coherence.

7 What certification implies for a task

7.1 A field-mediated communication channel

An entangled probe state does not specify a communication rate. A channel also requires an encoding, a causal transfer interaction, receiver access, and a decoder. Covariant qubit–field constructions and engineered high-capacity scalar-field protocols

provide concrete realizations under specified assumptions [95, 96]. They are not consequences of a positive harvesting witness alone. Quantum capacity further involves optimization over inputs and regularization over repeated channel uses [97, 98]. The admissible energy, timing and receiver operations must therefore be matched before capacities can be compared.

This separates two roles of the field. Spacelike probe interactions may transfer preexisting correlations, while sending a chosen message requires a causal input–output dependence. Conditional records or correlations shared with another receiver do not establish an individually controllable signal outside the sender's causal future. A task claim must describe the full map accessible to its users.

7.2 Parameter inference and identifiability

The same probe state can support different estimation tasks. For an actual readout distribution $p(x|\boldsymbol{\theta})$, the classical Fisher matrix is

$$F_{ij} = \sum_x p(x|\boldsymbol{\theta})\partial_i \ln p(x|\boldsymbol{\theta})\partial_j \ln p(x|\boldsymbol{\theta}), \tag{33}$$

where $\boldsymbol{\theta}$ is the parameter vector and a sum is replaced by an integral for continuous outcomes. For a single parameter, the quantum Fisher information is the maximum over measurements. Multiparameter bounds need not be simultaneously attainable; the allowed readout and parameter model remain essential [99, 100].

For a parameter of interest θ and nuisance vector $\boldsymbol{\nu}$, a regular, locally identifiable model with invertible $F_{\nu\nu}$ gives

$$F_{\mathrm{eff}} = F_{\theta\theta} - F_{\theta\nu}F_{\nu\nu}^{-1}F_{\nu\theta}, \qquad \mathrm{Var}(\widehat{\theta}) \geq \frac{1}{NF_{\mathrm{eff}}}. \tag{34}$$

Here N is the number of independent repetitions and the variance bound applies to unbiased estimators under the usual regularity conditions. When the readout depends on several unknown parameters only through a single effective temperature, its Fisher matrix has rank at most one: writing the distribution as $p(x|T_{\mathrm{eff}})$ makes each score proportional to $\partial_i T_{\mathrm{eff}}$. A nonzero sensitivity to one parameter therefore does not establish joint identifiability. Detector-thermometry and de Sitter studies supply model-specific examples of parameter-dependent precision [101, 102].

Preparation, interaction and readout also consume time. In a sequential protocol the repetition budget obeys $N \leq t_{\mathrm{tot}}/(t_{\mathrm{prep}} + t_{\mathrm{int}} + t_{\mathrm{read}})$, with the symbols denoting the total available time and the respective per-cycle durations. Postselection must include unsuccessful cycles. Resource certification and an attainable task advantage are thus related claims whose cost and identifiability premises remain independent.

8 Conclusions

A resource claim based on a local quantum-field measurement is an inference across several descriptions. Causal measurement theory constrains the operations; the interaction and complete spacetime support determine the channel to the probes; the

resource scenario determines what property is tested; and the calibrated record determines whether that property is certified. Agreement at one stage does not establish agreement at the others.

The examples give concrete consequences. Equal outcome probabilities permit different local back-action and different retained resources. Data processing bounds detector correlations only after the field algebras, product assumptions and localization have been fixed; it gives no positive harvesting yield by itself. Residual negativity can be maximal on a biseparable mixed state, while selected entanglement invariance can follow exactly from an initial state and a local channel without identifying a specifically relativistic mechanism. Contextuality claims can also differ because their preparation and measurement scenarios differ.

The distinction between prediction, certification and attribution is especially important for small signals. A truncated dynamical model needs its own error control. A directly measured witness needs calibration and sampling control. Attribution to the field additionally needs the preparation and causal premises of the harvesting protocol. Composite observables and noncompact profiles make these requirements more specific, rather than removing them. The resulting account connects foundational questions about local measurement with operational resource studies while keeping each conclusion at the level supported by its assumptions and evidence.

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Data and code availability

Analytic derivations, the retained numerical datasets, and Python verification scripts are supplied with the supplementary material and reproducibility archive. The scripts check specified finite-state and integration examples; they do not validate every cited result or omitted perturbative order. No experimental datasets were collected or analyzed.

Author contributions

Zhiming Huang is the sole author and is responsible for the submitted manuscript.

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